Final UML Diagrams

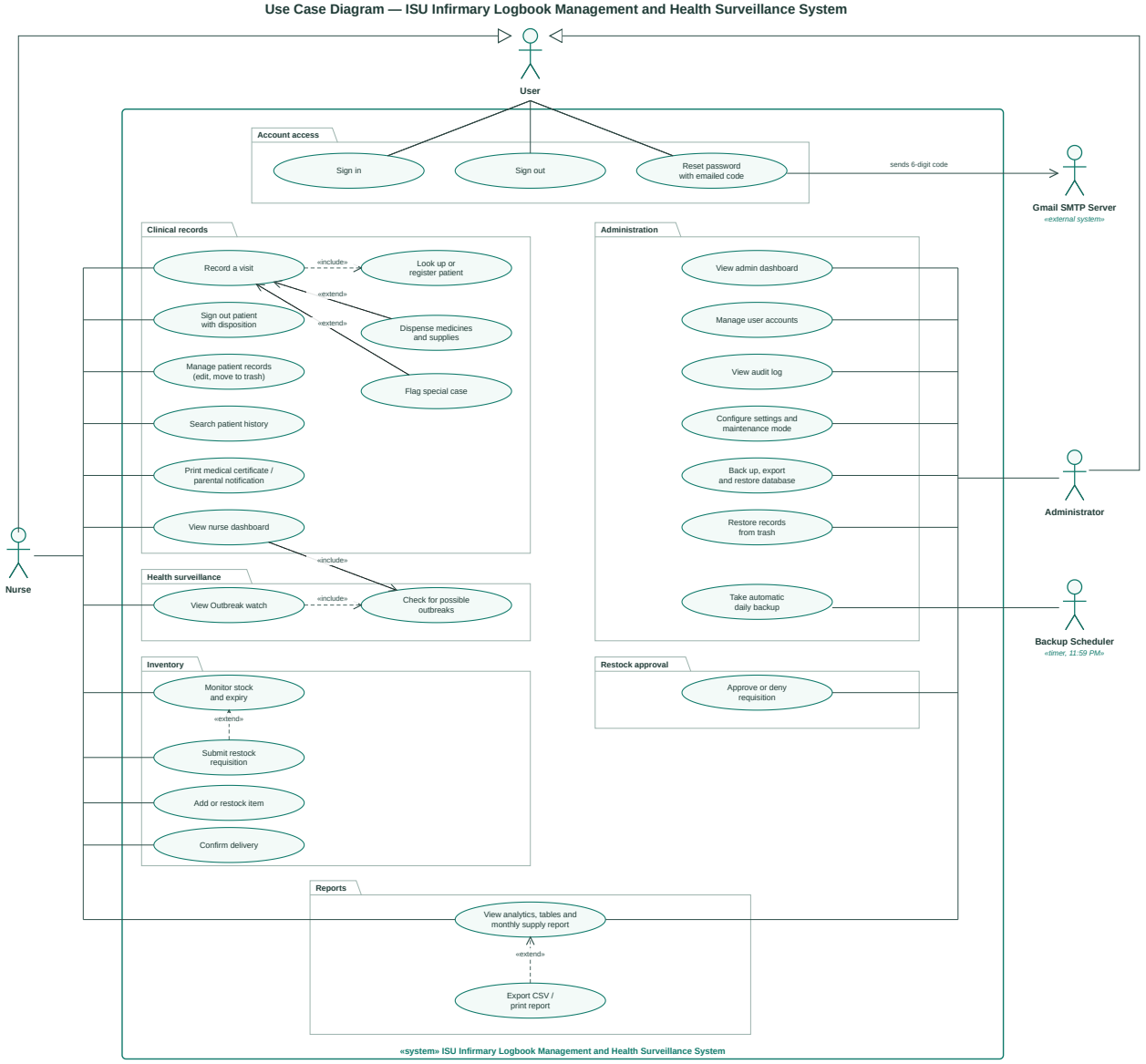
ISU Infirmary Logbook Management and Health Surveillance System

BSCS 3-1 · Gaffud, Cyrille Jef N. · Nicolas, Dj Jaziel P.

7 diagrams, drawn from the final system (GitHub commit d0090cf, November 2026)

1. Use Case Diagram

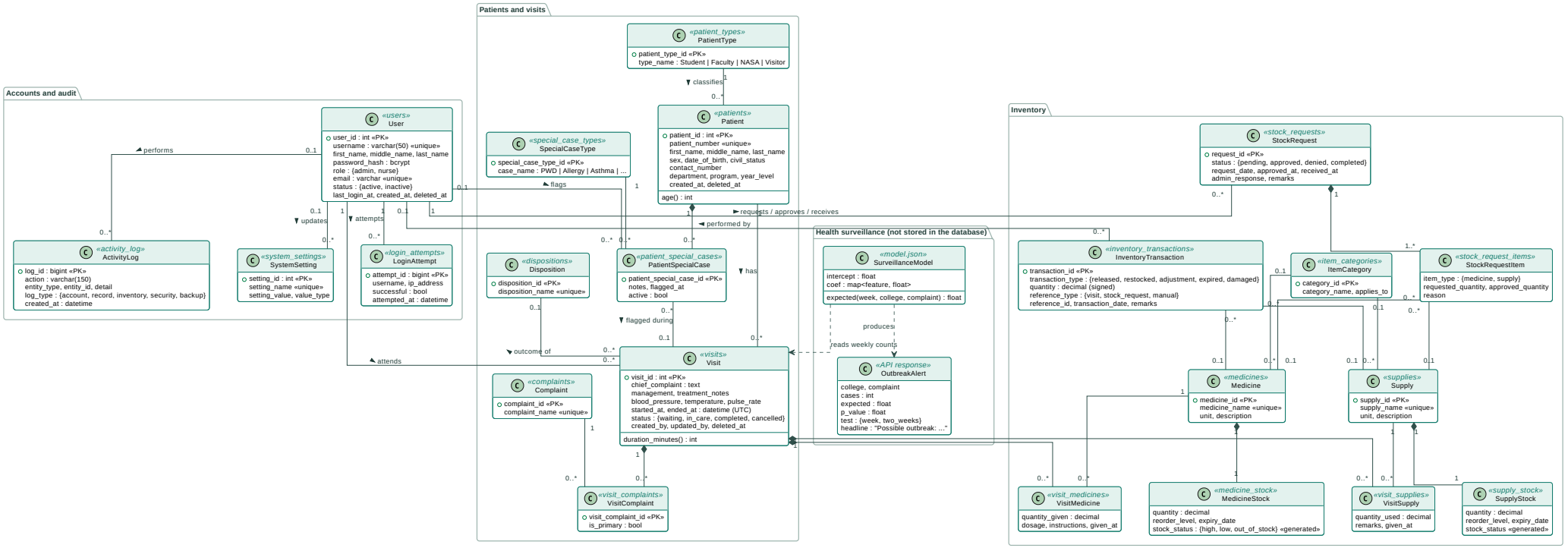
Who uses the system and what they can do. Nurse and Administrator both specialise the general User (sign in, sign out, reset password by emailed code through Gmail). The Nurse records visits, patients, special cases, stock and requisitions, and sees outbreak alerts; the Administrator approves requisitions and runs accounts, audit log, settings, backups and trash. Both view Reports. A timer takes the automatic backup at 11:59 PM.



2. Class Diagram

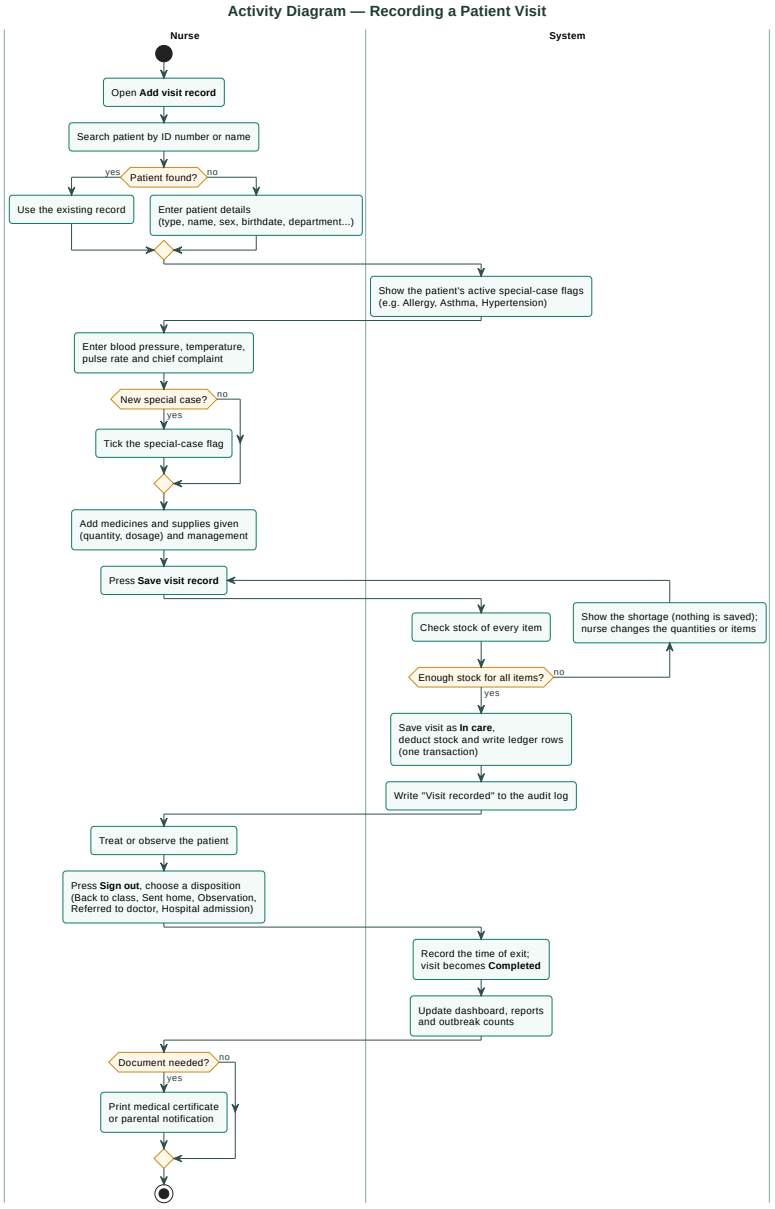
The 22 database tables as classes with their key attributes and multiplicities, grouped into Accounts and audit, Patients and visits, and Inventory. A Visit owns its complaint tags, medicines and supplies; every stock change is an InventoryTransaction. The health-surveillance model (model.json) reads weekly visit counts and produces outbreak alerts; it is not stored in the database.

Class Diagram — ISU Infirmary Logbook Management and Health Surveillance System
(the 22 database tables, plus the health-surveillance model; PK = primary key)



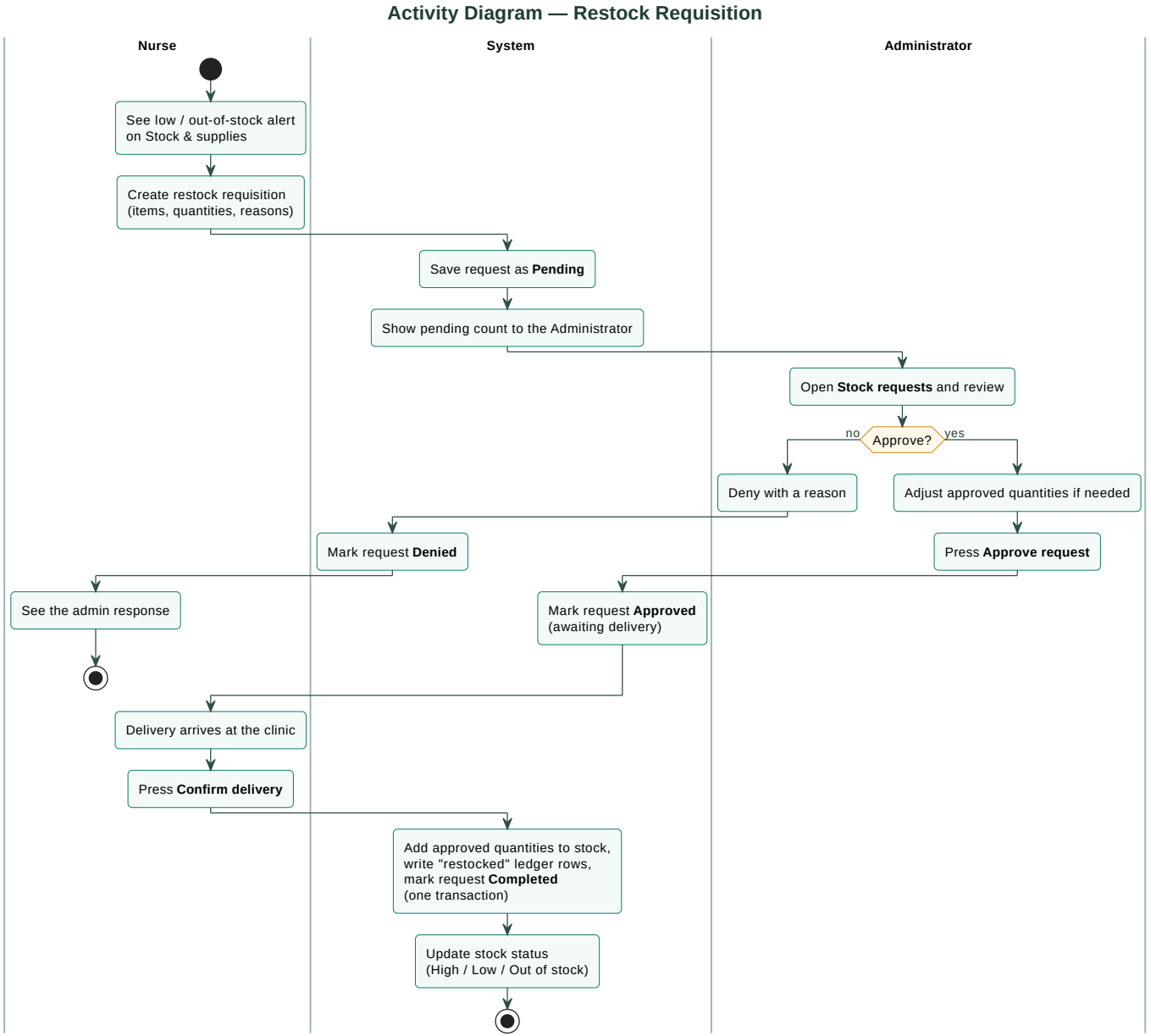
3. Activity Diagram — Recording a Patient Visit

From patient lookup to sign-out. When stock is short the system saves nothing and the nurse changes the items; otherwise the visit, stock deductions and ledger rows are saved together, and signing out completes the visit.



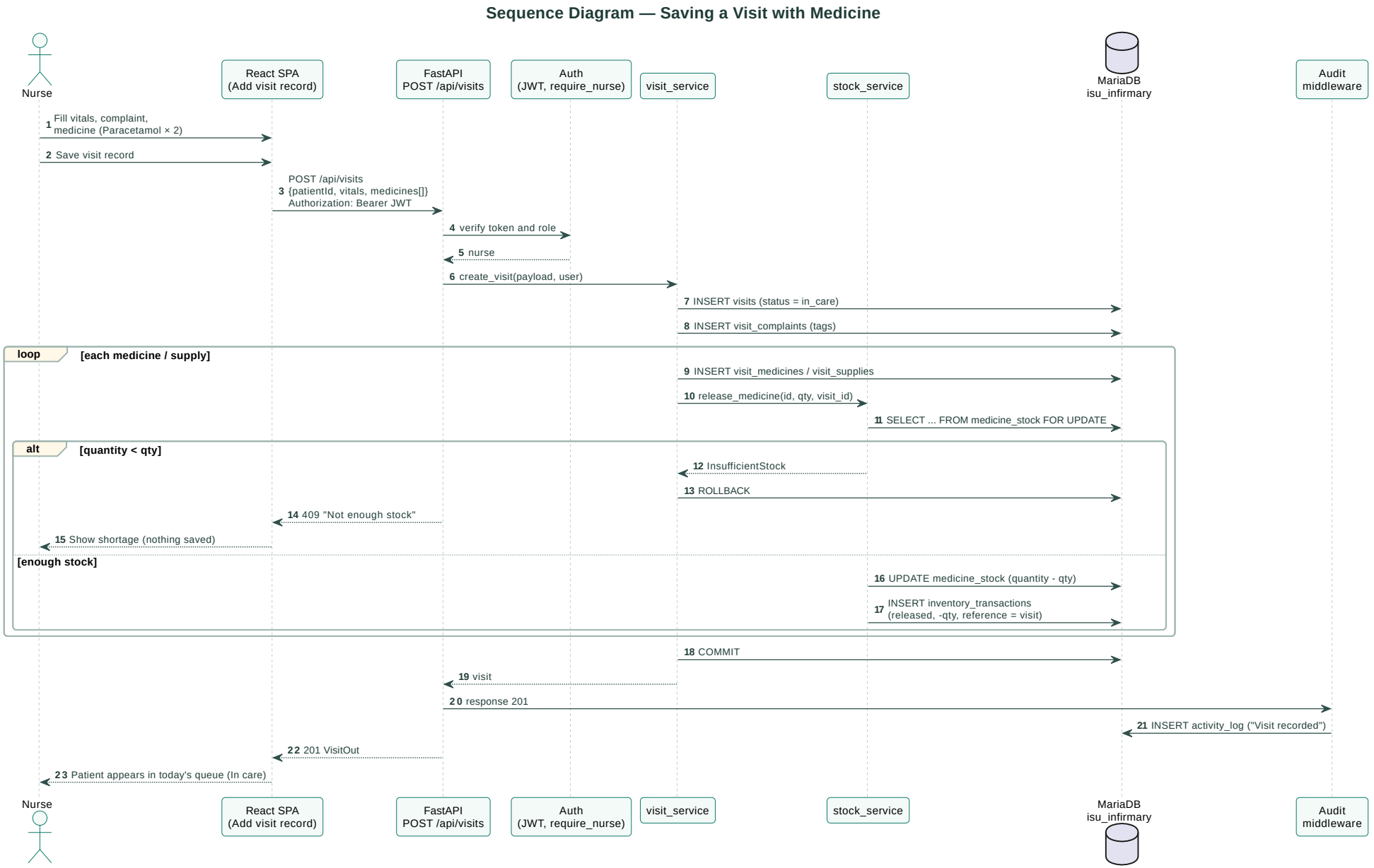
4. Activity Diagram — Restock Requisition

Nurse requests stock, Administrator approves (possibly with changed quantities) or denies, and the Nurse confirms the delivery, which adds the approved quantities to stock in one transaction.



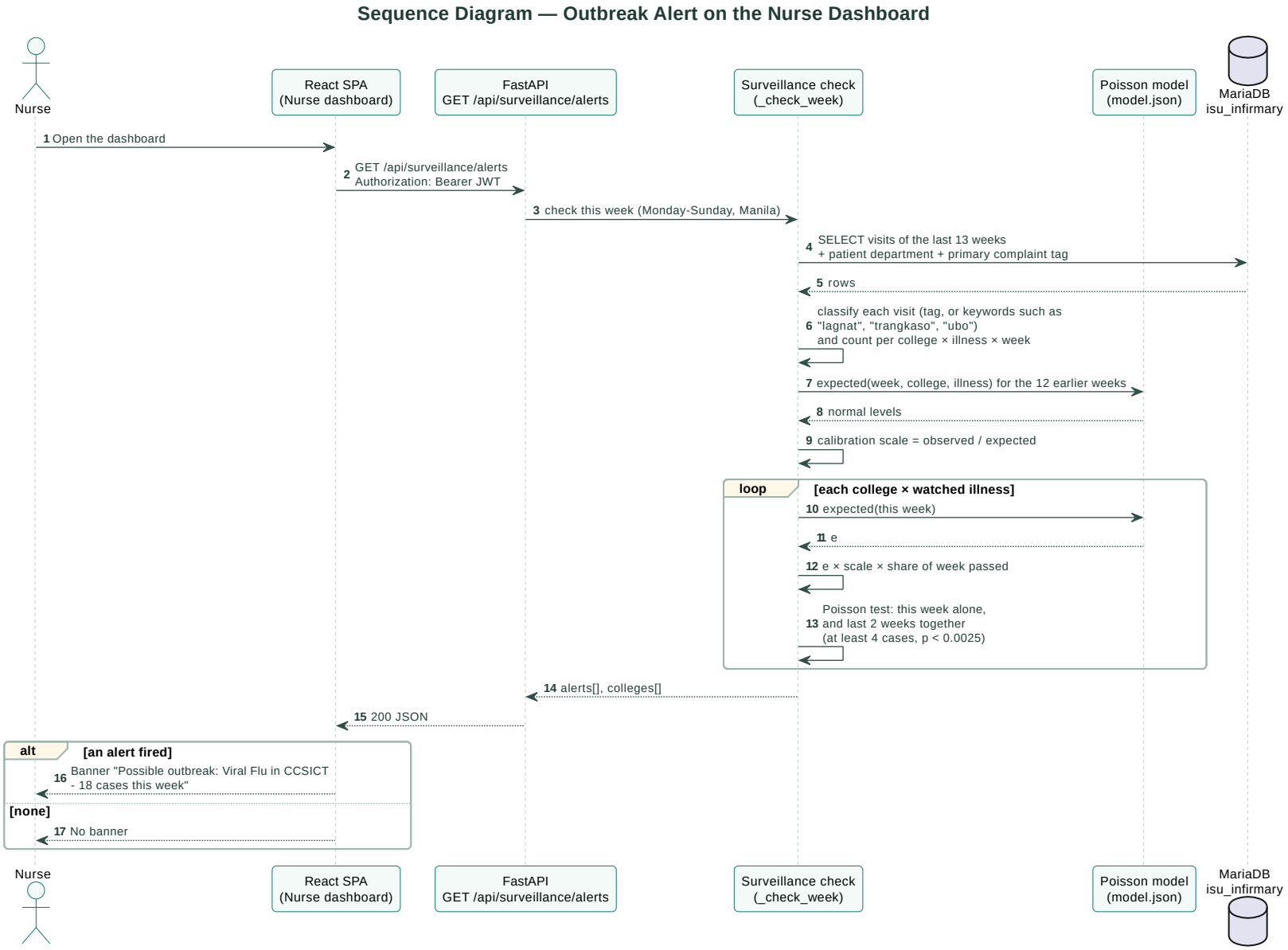
5. Sequence Diagram — Saving a Visit with Medicine

The messages between the React frontend, the FastAPI route, the visit and stock services and MariaDB, including the row lock on the stock, the 409 rollback when stock is short, and the audit-log entry.



6. Sequence Diagram — Outbreak Alert on the Nurse Dashboard

How the dashboard asks for this week's alerts: weekly counts per college and illness, the Poisson model's expected level calibrated to the clinic's last 12 weeks, and the alert rule (at least 4 cases, p < 0.0025).



7. Deployment Diagram

Where the software runs: one infirmary server PC with XAMPP MariaDB, the Python FastAPI backend and the Node/Vite frontend server; nurse workstations on the UHS LAN open port 5173 in a browser; only password-reset email leaves the LAN, through Gmail SMTP. Full explanation: 7_Deployment_Diagram_Explanation.docx.

UML Deployment Diagram — ISU Infirmary Logbook Management and Health Surveillance System

